Completion and Cyclic Ordering Problems for Oriented Graphs

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Abstract

For a class $\mathcal{C}$ of oriented graphs, the orientation completion problem asks whether a partially oriented graph can be completed to a member of $\mathcal{C}$ by orienting its unoriented edges. In this paper we resolve four open problems of Bang-Jensen, Huang and Zhu on orientation completions, acyclic linkage completions, and cyclic orderings of oriented graphs.

We first prove that the orientation completion problem for acyclic in-tournaments is polynomial-time solvable. The proof is based on a characterization in terms of perfect elimination orderings compatible with the prescribed arcs. We then prove that the acyclic π-linkage completion problem is NP-complete for every fixed $k \geq 1$, where k is the number of prescribed terminal pairs. The hardness already holds for $k = 1$, even when the prescribed digraph is acyclic.

We next consider nice cyclic orderings. We prove that deciding whether an oriented graph admits a nice cyclic ordering is NP-complete. We also construct an eight-vertex oriented graph which admits a nice cyclic ordering but admits no excellent cyclic ordering. Finally, we identify two tractable cases: every acyclic oriented graph admits a nice cyclic ordering, and on tournaments, nice cyclic orderings coincide with round orderings and hence can be recognized in polynomial time.

Keywords: Orientation completion; partially oriented graph; acyclic in-tournament; directed linkage; cyclic ordering; NP-completeness

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1 Introduction

1.1. Orientation completion and acyclic in-tournaments. For a fixed class $\mathcal{C}$ of oriented graphs, the *orientation completion problem* asks whether a given partially oriented graph can be completed to an oriented graph in $\mathcal{C}$. A partially oriented graph, or pog, is a triple $P = (V, E \cup A)$, where E is a set of unordered pairs of distinct vertices and A is the arc set of an oriented graph on V, such that no pair of vertices is joined both by an edge of E and by an arc of A. A completion of P is obtained by orienting every edge in E, while keeping every arc in A fixed.

Orientation completion problems form a common framework for recognition and extension problems. If $A = \emptyset$, one obtains the ordinary problem of deciding whether a graph admits an orientation in the prescribed class. If some arcs are already prescribed, one obtains an extension

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problem: the task is to decide whether a partial orientation can be extended without destroying the desired structure. Transitive orientations are closely connected with comparability graphs, while chordal graphs are characterized by perfect elimination orderings [20, 21]. Related orientation and representation algorithms for proper interval and proper circular-arc graphs were developed in [7, 10]. Partial representation extension for proper and unit interval graphs was studied in [16].

The systematic study of orientation completion problems was initiated by Bang-Jensen, Huang and Zhu [3]. They considered many natural classes, including local tournaments, acyclic local tournaments, locally transitive local tournaments, in-tournaments, connectivity classes, and directed cycle-factor classes, obtaining both polynomial time algorithms and NP-completeness results. Subsequent work has further developed the obstruction and complexity theory of orientation completion, including obstructions for acyclic local tournament completions [11], local tournament completion obstructions [12, 13], and a forbidden-tournament dichotomy of Bodirsky and Guzmán-Pro [5].

This paper studies three topics arising from the problems of Bang-Jensen, Huang and Zhu [3]: acyclic in-tournament completions, linkage completions in the acyclic setting, and cyclic orderings of oriented graphs. We resolve three of the open problems in their paper and also prove hardness for a natural acyclic analogue of their local-tournament linkage-completion problem.

We first consider acyclic in-tournaments. A digraph is an *in-tournament* if the set of in-neighbors of every vertex induces a tournament. In-tournaments were introduced and studied by Bang-Jensen, Huang and Prisner [4]. Bang-Jensen, Huang and Zhu proved that completion to in-tournaments is polynomial time solvable [3]. They also recalled the characterization that a graph admits an orientation as an acyclic in-tournament if and only if it is chordal [4]. Thus, in the acyclic case, the completion problem may be viewed as a constrained version of chordal elimination, where the prescribed arcs impose additional restrictions on a perfect elimination ordering. Bang-Jensen, Huang and Zhu left open the complexity of this completion problem [3]. We prove that it is polynomial time solvable.

Theorem 1.1. *The orientation completion problem for the class of acyclic in-tournaments is polynomial time solvable.*

1.2. Acyclic linkage completions. The second part of the paper concerns linkage completions. Let $\pi = \{(s_1, t_1), \dots, (s_k, t_k)\}$ be a set of ordered terminal pairs. A π-linkage is a collection of pairwise vertex-disjoint directed paths $R_1, \dots, R_k$, where R_i goes from s_i to t_i. Linkage problems are among the central topics of graph algorithms. In undirected graphs, the fixed k-disjoint paths problem is polynomial by the Robertson–Seymour linkage algorithm [19]. In directed graphs the situation is very different: the directed two-linkage problem is NP-complete in general digraphs [8, 17]. Moreover, arbitrarily high connectivity does not force a digraph to be 2-linked [22]. On the positive side, Chudnovsky, Scott and Seymour proved that the fixed k directed linkage problem is polynomial for tournaments, and more generally for semicomplete digraphs [6].

Bang-Jensen, Huang and Zhu asked for the complexity of the local-tournament-π-linkage completion problem when $k \geq 2$ is fixed. In this paper we consider the corresponding acyclic completion variant. Thus we ask whether the unoriented edges of a partially oriented graph can be oriented so that the resulting oriented graph is acyclic and contains a prescribed π-linkage. When $A = \emptyset$ and k is fixed, the problem reduces to the undirected k-disjoint paths problem and is polynomial by the Robertson–Seymour linkage algorithm. When the input is already an acyclic oriented graph and k is fixed, the directed linkage problem can also be solved in polynomial time by dynamic programming over a topological ordering. We show that the mixed completion setting is nevertheless hard.

Theorem 1.2. *For every fixed integer $k \geq 1$, the acyclic π-linkage completion problem with k terminal pairs is* NP*-complete. This remains true even when the prescribed digraph (V, A) is acyclic, and already for $k = 1$. Consequently, the unrestricted version, in which k is part of the input, is also* NP*-complete.*

1.3. Nice cyclic orderings. The final part of the paper concerns cyclic orderings. A cyclic ordering of an oriented graph is closely related to the structure of local tournaments and round digraphs. Round orderings characterize connected locally transitive local tournaments, and locally transitive tournaments can be described through highly regular tournaments and substitutions of transitive tournaments, see [1, 14, 15, 18]. In their study of orientation completion, Bang-Jensen, Huang and Zhu introduced excellent cyclic orderings and used them to prove hardness results for completions to locally transitive tournaments.

For distinct vertices x, y, z in a cyclic ordering O, write $\mathrm{Cyc}_O(x, y, z)$ if, starting from x and moving clockwise around O, one meets y before z. A cyclic ordering $O = v_1, v_2, \ldots, v_n, v_1$ of an oriented graph D is called *nice* if there are no vertices v_k, v_i, v_j occurring in this cyclic order such that both $v_i \to v_k$ and $v_j \to v_i$ are arcs of D. It is called *excellent* if it has no two arcs $v_i \to v_j$ and $v_s \to v_t$ whose endpoints occur in the forbidden cyclic pattern v_i, v_t, v_s, v_j, with the usual endpoint coincidences allowed.

The key observation behind our hardness proof is that niceness can be tested on directed paths of length two: a cyclic ordering is nice precisely when every directed path $x \to y \to z$ appears in the cyclic order $\mathrm{Cyc}_O(x, y, z)$.

Every excellent cyclic ordering is nice. Bang-Jensen, Huang and Zhu asked for the complexity of deciding whether an oriented graph has a nice cyclic ordering. They also asked whether the converse implication holds, namely whether every oriented graph with a nice cyclic ordering must have an excellent cyclic ordering. The classical Cyclic Ordering problem, proved NP-complete by Galil and Megiddo [9], provides the starting point for our answer to the first question.

Theorem 1.3. *It is* NP*-complete to decide whether a given oriented graph has a nice cyclic ordering.*

We also record two simple positive cases. First, every acyclic oriented graph admits a nice cyclic ordering. Second, on tournaments, niceness coincides with the standard notion of roundness. Since round orderings can be recognized and found in polynomial time [3, Theorem 3], this yields a polynomial time algorithm for nice cyclic ordering recognition on tournaments.

Theorem 1.4. *Let T be a tournament and let O be a cyclic ordering of $V(T)$. Then O is nice if and only if O is round. Consequently, nice cyclic ordering recognition is polynomial time solvable on tournaments.*

We also answer the last problem in [3] negatively.

Theorem 1.5. *There exists an oriented graph which has a nice cyclic ordering but has no excellent cyclic ordering.*

Organization of the paper. In Section 2 we prove Theorem 1.1. In Section 3 we prove Theorem 1.2. In Section 4 we study nice cyclic orderings: we prove Theorem 1.3, record two tractable restricted cases including Theorem 1.4, and give the eight-vertex example separating nice and excellent cyclic orderings. Finally, we end with some remarks in Section 5.

Notation. For notation not defined in this paper, we refer the reader to [2]. All graphs and digraphs in this paper are finite. Graphs have no loops or multiple edges, and oriented graphs have no loops and no pair of opposite arcs. If $P = (V, E \cup A)$ is a partially oriented graph, then $G(P)$ denotes its underlying graph; two vertices are adjacent in $G(P)$ precisely when they are joined in P either by an edge or by an arc in one of the two directions.

An ordering $\tau = (v_1, \dots, v_n)$ of the vertices of a graph G is a *perfect elimination ordering*, or *PEO*, if for every i, the set of neighbors of v_i appearing after v_i in τ is a clique. A vertex v of a graph H is *simplicial* if its whole neighborhood in H is a clique.

For a cyclic ordering O and two distinct vertices a, b, we write $(a, b)_O$ for the open clockwise interval from a to b. Thus $x \in (a, b)_O$ means $\mathrm{Cyc}_O(a, x, b)$. For pairwise distinct vertices $x_1, \dots, x_m$, the notation $\mathrm{Cyc}_O(x_1, \dots, x_m)$ means that, starting from x_1 and moving clockwise around O, the vertices $x_2, \dots, x_m$ are encountered in this order.

2 Completion to acyclic in-tournaments

We prove Theorem 1.1 by reducing the problem to finding a perfect elimination ordering compatible with the prescribed arcs.

Let $P = (V, E \cup A)$ be a partially oriented graph and let $G = G(P)$. For each arc $xy \in A$, impose the precedence constraint $y \prec x$. Let R_A be the directed graph of these constraints, where an arc yx represents the constraint $y \prec x$. An ordering τ of V is called R_A-compatible if every constraint $y \prec x$ places y before x in τ.

Lemma 2.1. *The partially oriented graph P has a completion to an acyclic in-tournament if and only if $G(P)$ has an R_A-compatible perfect elimination ordering.*

Proof. Suppose first that P has a completion D which is an acyclic in-tournament. Let $\sigma = (u_1, \dots, u_n)$ be a topological ordering of D, so every arc of D is directed from earlier to later in σ. Let $\tau = (u_n, \dots, u_1)$ be the reverse order. Fix a vertex v. The neighbors of v appearing after v in τ are exactly the neighbors of v appearing before v in σ, and these are precisely the in-neighbors of v in D. Since D is an in-tournament, these vertices form a clique in $G(P)$. Hence τ is a PEO of $G(P)$. If $xy \in A$, then x appears before y in σ, so y appears before x in τ. Therefore τ is R_A-compatible.

Conversely, suppose that $G(P)$ has an R_A-compatible PEO $\tau = (v_1, \dots, v_n)$. Let $\sigma = (v_n, \dots, v_1)$ be the reverse order, retain every arc in A, and orient each edge of E from its earlier endpoint to its later endpoint in σ. Equivalently, for each edge $v_iv_j \in E$ with $i < j$, orient it as $v_j \to v_i$. The resulting orientation is acyclic. It preserves every arc in A, because $xy \in A$ gives the constraint $y \prec x$, so x is before y in σ. Finally, for every vertex v, its in-neighbors in the constructed orientation are exactly its neighbors appearing after v in τ. They form a clique by the PEO property, and the linear order σ orients this clique as a transitive tournament. Thus the completion is an acyclic in-tournament. $\square$

We now give a greedy recognition algorithm. During the algorithm, H denotes the current induced subgraph of $G(P)$, and R denotes the current induced subdigraph of R_A. A current vertex is called admissible if it is simplicial in H and has indegree zero in R.

Algorithm 1 Greedy compatible elimination

Input: A partially oriented graph $P = (V, E \cup A)$.

Output: Either an R_A-compatible PEO of $G(P)$, or the conclusion that none exists.

Construct $G = G(P)$ and the constraint digraph R_A.
Set $H := G$, $R := R_A$, and $\tau := \emptyset$.
while $V(H) \neq \emptyset$ **do**
    Find an admissible vertex v of the current pair (H, R).
    if no admissible vertex exists **then**
        reject.
    end if
    append v to τ, and delete v from both H and R.
end while
accept and output τ.

We use two elementary facts to justify the greedy choices.

Lemma 2.2. *If τ is a PEO of a graph H and $S \subseteq V(H)$, then the restriction of τ to S is a PEO of $H[S]$.*

Proof. For each $x \in S$, the later neighbors of x in the restricted order are a subset of the later neighbors of x in τ. Since the latter set is a clique in H, the former set is a clique in $H[S]$. $\square$

Lemma 2.3. *Suppose that the current pair (H, R) has an R-compatible PEO. If v is admissible, then $(H - v, R - v)$ has an $(R - v)$-compatible PEO. Moreover, v can be placed first before any $(R - v)$-compatible PEO of $H - v$ without destroying either compatibility or the PEO property.*

Proof. Let π be an R-compatible PEO of H. By Lemma 2.2, the restriction of π to $V(H) \setminus \{v\}$ is a PEO of $H - v$, and it clearly respects all constraints of $R - v$. Thus the remaining instance is feasible.

If ρ is any $(R - v)$-compatible PEO of $H - v$, then placing the simplicial vertex v first and then following ρ gives a PEO of H. Since v has indegree zero in R, no current constraint requires another vertex to appear before v. Hence the resulting order is R-compatible. $\square$

The next lemma gives the converse obstruction: if no admissible vertex exists, then no compatible elimination order can start.

Lemma 2.4. *If the current pair (H, R) has no admissible vertex, then it has no R-compatible PEO.*

Proof. If an R-compatible PEO existed, let v be its first vertex. Since v is first, all of its current neighbors are later neighbors, so they must form a clique. Thus v is simplicial in H. Also, no constraint can require another current vertex to appear before v, so v has indegree zero in R. This makes v admissible, a contradiction. $\square$

Proof of Theorem 1.1. By Lemma 2.1, it is enough to decide whether $G(P)$ has an R_A-compatible PEO. If algorithm 1 accepts, then every chosen vertex was simplicial in the graph remaining at the time it was chosen, so the output order is a PEO. Since a vertex is chosen only when it has indegree zero in the remaining constraint digraph, no vertex is placed before one of its required predecessors. Hence the output order is R_A-compatible, and Lemma 2.1 gives an acyclic in-tournament completion.

Conversely, suppose that such a completion exists. Again by Lemma 2.1, the initial pair has a compatible PEO. As long as the current pair is feasible, Lemma 2.4 guarantees the existence of an admissible vertex, and Lemma 2.3 shows that deleting any admissible vertex preserves feasibility. Therefore the algorithm cannot reject a yes-instance. Let $n = |V(P)|$. There are at most n iterations. In each iteration, at most n vertices are examined, and testing whether one vertex is simplicial takes $O(n^2)$ time using an adjacency matrix. The indegrees in the constraint digraph can be updated in $O(n)$ time after each deletion. Consequently, a direct implementation runs in $O(n^4)$ time. $\square$

3 Acyclic linkage completion

Let $\pi = \{(s_1, t_1), \dots, (s_k, t_k)\}$ be a set of ordered terminal pairs in a digraph, where the $2k$ terminal vertices $s_1, t_1, \dots, s_k, t_k$ are pairwise distinct. Throughout this section, all paths are vertex-simple. A π-linkage is a collection of pairwise vertex-disjoint directed paths $R_1, \dots, R_k$, where R_i starts at s_i and ends at t_i. The acyclic π-linkage completion problem asks whether a partially oriented graph P can be completed to an acyclic oriented graph containing a π-linkage.

We shall use the following elementary extension lemma repeatedly.

Lemma 3.1. *Let $P = (V, E \cup A)$ be a partially oriented graph. Let F be a set of arcs obtained by orienting some edges of E. Then there exists an acyclic completion of P containing every arc of F if and only if the digraph $(V, A \cup F)$ is acyclic.*

Proof. The necessity is immediate, because every subdigraph of an acyclic digraph is acyclic. Conversely, suppose $(V, A \cup F)$ is acyclic and take a topological ordering σ of it. Orient every remaining unoriented edge from its earlier endpoint to its later endpoint in σ. Then every arc of the completed orientation goes forward in σ, so the completion is acyclic and contains all arcs of $A \cup F$. $\square$

We first prove NP-hardness for one terminal pair by a reduction from 3-Sat. The construction is a chain of variable and clause diamonds: a directed s-t path will choose one branch in each diamond, while the prescribed blocking arcs encode falsified literals.

Let Φ be a 3-CNF formula with variables $x_1, \dots, x_n$ and clauses $C_1, \dots, C_m$. Write the three literal occurrences in C_j as $\lambda_{j,1}, \lambda_{j,2}, \lambda_{j,3}$.

Construct a partially oriented graph $P_\Phi = (V, E \cup A)$ as follows. Introduce articulation vertices $q_0, q_1, \dots, q_{n+m}$, and put $s = q_0$ and $t = q_{n+m}$. For every variable x_i, introduce two vertices T_i and F_i, and add the four unoriented edges $q_{i-1}T_i$, T_iq_i, $q_{i-1}F_i$ and F_iq_i. For every clause C_j and every $r \in \{1, 2, 3\}$, introduce a vertex $c_{j,r}$, and add the two unoriented edges $q_{n+j-1}c_{j,r}$ and $c_{j,r}q_{n+j}$.

It remains to add the prescribed blocking arcs. For each literal occurrence $\lambda_{j,r}$, if $\lambda_{j,r} = x_i$, add the arc $c_{j,r} \to F_i$; if $\lambda_{j,r} = \neg x_i$, add the arc $c_{j,r} \to T_i$. The arc set A is acyclic, since every blocking arc goes from a clause-literal vertex to a variable-choice vertex, and no blocking arc leaves a variable-choice vertex.

Lemma 3.2. *Let D be any completion of P_Φ, and let R be a directed path from $s = q_0$ to $t = q_{n+m}$ in D. Then R uses no blocking arc. Consequently, as an undirected path in the chain of diamonds, R has the form*

$$q_0, X_1, q_1, X_2, q_2, \dots, X_n, q_n, Y_1, q_{n+1}, Y_2, q_{n+2}, \dots, Y_m, q_{n+m},$$

where $X_i \in \{T_i, F_i\}$ for every i and $Y_j \in \{c_{j,1}, c_{j,2}, c_{j,3}\}$ for every j.

Proof. Suppose for a contradiction that R uses a blocking arc, and choose the first blocking arc on R. It has the form $c_{j,r} \to X_i$, where $X_i \in \{T_i, F_i\}$. Before using this arc, the path uses only edges of the diamond chain. All variable blocks occur before all clause blocks in the chain. Since $c_{j,r}$ lies in a clause block, this initial segment has already passed through both q_{i-1} and q_i. After entering X_i through the blocking arc, the path can only continue to q_{i-1} or q_i, because no blocking arc leaves a variable-choice vertex and X_i has no other incident edge in the diamond chain. Both possible next vertices have already appeared on R, contradicting that R is a path. Thus no blocking arc is used.

Once blocking arcs are excluded, R is a simple undirected q_0-q_{n+m} path in the chain of diamonds. Each articulation vertex q_ℓ, with $1 \le \ell \le n + m - 1$, separates q_0 from q_{n+m} in this chain. Hence R passes through the articulation vertices in their natural order and chooses exactly one internal branch vertex in each diamond block. This gives the stated form. $\square$

Lemma 3.3. *If Φ is satisfiable, then P_Φ has an acyclic completion containing a directed path from s to t.*

Proof. Let α be a satisfying assignment. For each variable x_i, choose $X_i = T_i$ if $\alpha(x_i) =$ true, and choose $X_i = F_i$ otherwise. For each clause C_j, choose one literal occurrence λ_{j,r_j} that is true under α, and put $Y_j = c_{j,r_j}$. Let R be the path through the chosen vertices in the order displayed in Lemma 3.2, and orient every edge of R forward along this order. Denote this set of oriented path edges by F_R.

We claim that the digraph $(V, A \cup F_R)$ is acyclic. Suppose, to the contrary, that it contains a directed cycle C. Since the arcs of F_R form a directed path, the cycle C must contain at least one prescribed blocking arc. Every blocking arc has the form $c_{j,r} \to Z_i$, where $Z_i \in \{T_i, F_i\}$.

We first observe that the tail $c_{j,r}$ of any blocking arc belonging to C must be the clause vertex selected by R in the jth clause block. Indeed, if $c_{j,r} \neq Y_j$, then no arc of F_R is incident with $c_{j,r}$. Moreover, no blocking arc enters a clause vertex. Thus $c_{j,r}$ has indegree zero in $(V, A \cup F_R)$, and so it cannot lie on a directed cycle.

Similarly, the head Z_i of any blocking arc belonging to C must be the variable vertex selected by R in the ith variable block. Indeed, if $Z_i \neq X_i$, then no arc of F_R is incident with Z_i, and no blocking arc leaves a variable vertex. Thus Z_i has outdegree zero in $(V, A \cup F_R)$, and again it cannot lie on a directed cycle.

Consequently, C contains a blocking arc of the form $Y_j \to X_i$, where both Y_j and X_i are selected by R. This is impossible. The vertex Y_j represents a literal that is true under α, whereas its blocking arc is directed to the variable-choice vertex that makes this literal false. Hence the head of the blocking arc leaving Y_j is the unselected vertex in $\{T_i, F_i\}$, not X_i.

Therefore $(V, A \cup F_R)$ is acyclic. By Lemma 3.1, the orientation of the edges of R extends to an acyclic completion of P_Φ containing R as a directed path from s to t. $\square$

Lemma 3.4. *If P_Φ has an acyclic completion containing a directed path from s to t, then Φ is satisfiable.*

Proof. Let D be an acyclic completion of P_Φ, and let R be a directed s-t path in D. By Lemma 3.2, R chooses one vertex $X_i \in \{T_i, F_i\}$ in every variable block and one vertex $Y_j \in \{c_{j,1}, c_{j,2}, c_{j,3}\}$ in every clause block. Define a truth assignment by setting x_i true exactly when $X_i = T_i$.

Fix a clause C_j, and suppose $Y_j = c_{j,r}$. If the literal $\lambda_{j,r}$ were false under the assignment, then the construction would contain the blocking arc $Y_j \to X_i$, where X_i is the variable vertex selected by R. On the directed path R, the vertex X_i appears before Y_j, so R contains a directed subpath from X_i to Y_j. Together with the blocking arc $Y_j \to X_i$, this forms a directed cycle in D, contradicting the acyclicity of the completion. Therefore the selected literal in every clause is true, and Φ is satisfiable. $\square$

Proof of Theorem 1.2. The preceding two lemmas show that Φ is satisfiable if and only if P_Φ has an acyclic completion containing a directed path from s to t. This proves NP-hardness for $k = 1$, even though the prescribed arc set A of P_Φ is acyclic.

The problem is in NP: a certificate consists of an orientation of every edge and the displayed directed paths. One checks in polynomial time that the orientation preserves A, is acyclic, and contains pairwise vertex-disjoint paths linking the prescribed pairs.

For fixed $k \geq 2$, add new vertices $s_2, t_2, \ldots, s_k, t_k$, add the single prescribed arc $s_i \to t_i$ for each $i \in \{2, \ldots, k\}$, and add no other incident edge or arc at these new vertices. Use the terminal pairs $(s, t), (s_2, t_2), \ldots, (s_k, t_k)$. The added terminal paths are forced to be the single arcs $s_i \to t_i$, and they are isolated from the original construction. Thus the enlarged instance has an acyclic π-linkage completion if and only if the original $k = 1$ instance has an acyclic completion containing a directed s-t path. This proves NP-hardness for every fixed integer $k \geq 2$, and hence, together with the case $k = 1$, for every fixed integer $k \geq 1$.

Finally, consider the unrestricted version in which k is part of the input. Membership in NP follows from the same certificate argument. Moreover, this version contains as a restriction the already NP-hard case $k = 1$. Therefore the unrestricted version is also NP-complete. $\square$

4 Cyclic orderings

Throughout this section, excellent cyclic orderings are understood with the endpoint convention of Bang-Jensen, Huang and Zhu [3]: for two arcs $v_i \to v_j$ and $v_s \to v_t$, the forbidden cyclic pattern is v_i, v_t, v_s, v_j, allowing the endpoint coincidences $v_i = v_t$ or $v_s = v_j$. In particular, if $v_s = v_j$, the forbidden pattern means $\mathrm{Cyc}_O(v_i, v_t, v_j)$.

4.1 Nice cyclic orderings

We next prove Theorem 1.3. We first make the observation from the introduction precise.

Lemma 4.1. *Let D be an oriented graph and let O be a cyclic ordering of $V(D)$. Then O is nice if and only if every directed path $x \to y \to z$ in D satisfies* $\mathrm{Cyc}_O(x, y, z)$.

Proof. Suppose first that O is nice, and let $x \to y \to z$ be a directed path. Since D is oriented, the vertices x, y, z are distinct. If $\mathrm{Cyc}_O(x, y, z)$ failed, then $\mathrm{Cyc}_O(z, y, x)$ would hold. Taking $v_k = z$, $v_i = y$, and $v_j = x$, we obtain a forbidden nice triple, because $v_i \to v_k$ and $v_j \to v_i$ are precisely the arcs $y \to z$ and $x \to y$. This contradicts niceness.

Conversely, suppose every directed path $x \to y \to z$ satisfies $\mathrm{Cyc}_O(x, y, z)$. If O were not nice, there would be vertices v_k, v_i, v_j occurring in this cyclic order such that $v_i \to v_k$ and $v_j \to v_i$. Then $v_j \to v_i \to v_k$ is a directed path, so the assumption gives $\mathrm{Cyc}_O(v_j, v_i, v_k)$, contradicting the cyclic order $\mathrm{Cyc}_O(v_k, v_i, v_j)$. Hence O is nice. $\square$

Lemma 4.2. *Every excellent cyclic ordering of an oriented graph is nice.*

Proof. Let O be an excellent cyclic ordering, and let $x \to y \to z$ be a directed path. Suppose that $\mathrm{Cyc}_O(x, y, z)$ does not hold. Since x, y, z are distinct, it follows that $\mathrm{Cyc}_O(x, z, y)$ holds.

Consider the two arcs $x \to y$ and $y \to z$. In the definition of an excellent cyclic ordering, take $v_i = x, v_j = y, v_s = y, v_t = z$. Here the allowed endpoint coincidence $v_s = v_j = y$ occurs. Thus $\mathrm{Cyc}_O(x, z, y)$ is precisely the forbidden pattern for the two arcs $x \to y$ and $y \to z$, contradicting the excellence of O.

Hence every directed path $x \to y \to z$ satisfies $\mathrm{Cyc}_O(x, y, z)$. By Lemma 4.1, the cyclic ordering O is nice. $\square$

The following simple observation will be useful in the reduction.

Observation 4.3. *If O is a nice cyclic ordering of an oriented graph D, then the restriction of O to any induced subdigraph of D is nice.*

Proof. Every directed path of length two in D' is also a directed path of length two in D. Hence the restriction of O to $V(D')$ satisfies the condition in Lemma 4.1, and is therefore nice. $\square$

The source problem is Cyclic Ordering: given a finite set X and a family $\mathcal{C} \subseteq X^3$ of ordered triples of distinct elements, decide whether there is a cyclic ordering of X satisfying $\mathrm{Cyc}_O(a, b, c)$ for every $(a, b, c) \in \mathcal{C}$. This problem is NP-complete [9].

For every ordered triple (a, b, c), define an oriented gadget $G(a, b, c)$ as follows. Add three new vertices p, q, r, add the directed triangle $p \to q \to r \to p$, and add the six arcs

$$a \to q, \quad a \to r, \quad b \to r, \quad b \to p, \quad c \to p, \quad c \to q.$$

There are no other arcs in the gadget.

Lemma 4.4. *Let O_X be a cyclic ordering of the terminal set $\{a, b, c\}$. Then O_X extends to a nice cyclic ordering of $G(a, b, c)$ if and only if $\mathrm{Cyc}_{O_X}(a, b, c)$.*

Proof. Let O be a nice cyclic ordering of the gadget. The directed path $p \to q \to r$ gives $\mathrm{Cyc}_O(p, q, r)$ by Lemma 4.1. Thus the vertices p, q, r split the circle into the three intervals $(p, q)_O$, $(q, r)_O$, and $(r, p)_O$.

The directed paths $a \to q \to r$ and $a \to r \to p$ force $\mathrm{Cyc}_O(a, q, r)$ and $\mathrm{Cyc}_O(a, r, p)$. With $\mathrm{Cyc}_O(p, q, r)$, these two conditions place a in $(p, q)_O$. Indeed, placing a in either of the other two intervals violates one of the two conditions. Similarly, the paths $b \to r \to p$ and $b \to p \to q$ force $b \in (q, r)_O$, while the paths $c \to p \to q$ and $c \to q \to r$ force $c \in (r, p)_O$. Therefore the terminal vertices occur in the cyclic order a, b, c.

Conversely, suppose that $\mathrm{Cyc}_{O_X}(a, b, c)$. Insert q into $(a, b)_{O_X}$, r into $(b, c)_{O_X}$, and p into $(c, a)_{O_X}$, and denote the resulting cyclic ordering by O. Then $\mathrm{Cyc}_O(p, q, r)$, and the placements give $a \in (p, q)_O$, $b \in (q, r)_O$, and $c \in (r, p)_O$. The directed paths of length two in $G(a, b, c)$ are precisely

$$p \to q \to r, \quad q \to r \to p, \quad r \to p \to q,$$

$$a \to q \to r, \quad a \to r \to p, \quad b \to r \to p, \quad b \to p \to q, \quad c \to p \to q, \quad c \to q \to r.$$

Under the placement $q \in (a, b)_O$, $r \in (b, c)_O$, and $p \in (c, a)_O$, these respectively give

$$\mathrm{Cyc}_O(p, q, r), \quad \mathrm{Cyc}_O(q, r, p), \quad \mathrm{Cyc}_O(r, p, q),$$

$$\mathrm{Cyc}_O(a, q, r), \quad \mathrm{Cyc}_O(a, r, p), \quad \mathrm{Cyc}_O(b, r, p), \quad \mathrm{Cyc}_O(b, p, q), \quad \mathrm{Cyc}_O(c, p, q), \quad \mathrm{Cyc}_O(c, q, r).$$

Thus every directed path of length two satisfies the condition in Lemma 4.1. Hence the resulting cyclic ordering is nice. $\square$

Proof of Theorem 1.3. Given an instance $(X, \mathcal{C})$ of Cyclic Ordering, construct an oriented graph D as follows. Start with the vertex set X and no arcs between vertices of X. For every triple $\tau = (a, b, c) \in \mathcal{C}$, add a fresh copy of the gadget $G(a, b, c)$, with internal vertices p_τ, q_τ, r_τ. Different gadgets are vertex-disjoint outside their terminal vertices in X. The construction is polynomial.

If the cyclic ordering instance is satisfiable, take a satisfying cyclic order of X. For every $\tau = (a, b, c)$, place q_τ in (a, b), r_τ in (b, c), and p_τ in (c, a). These placements can be made independently. Since all arcs incident with vertices of X leave X, no directed path of length two has a vertex of X as its middle vertex. Also, no directed path of length two uses internal vertices from two different gadgets. Therefore the verification of niceness is local to each copy of $G(a, b, c)$. Hence every directed path of length two is contained in one gadget, possibly starting at a terminal, and Lemma 4.4 shows that all such paths have the required cyclic order. By Lemma 4.1, the whole ordering is nice.

Conversely, suppose D has a nice cyclic ordering. Restrict it to X. For each triple $\tau = (a, b, c) \in \mathcal{C}$, by Observation 4.3, the restriction to the vertices $a, b, c, p_\tau, q_\tau, r_\tau$ is a nice ordering of a copy of $G(a, b, c)$. By Lemma 4.4, the restricted order satisfies $\mathrm{Cyc}_O(a, b, c)$. Thus the restriction to X satisfies every triple in $\mathcal{C}$. This proves NP-hardness. Membership in NP follows from Lemma 4.1, since a proposed cyclic order can be checked by testing all directed paths of length two, of which there are at most $|V(D)|^3$. □

4.2 Two tractable restricted classes

We begin with the acyclic case. We shall use the following characterization of excellent cyclic orderings.

Lemma 4.5 ([3, Lemma 2])**.** *Let D be an oriented graph and let O be a cyclic ordering of $V(D)$. Then O is an excellent cyclic ordering of D if and only if D can be extended, by adding arcs, to a round local tournament for which O is a round ordering.*

Proposition 4.6. *Every topological ordering of an acyclic oriented graph, viewed as a cyclic ordering, is excellent and hence nice. Consequently, every acyclic oriented graph has a nice cyclic ordering.*

Proof. Let D be an acyclic oriented graph, and let $v_1, \ldots, v_n$ be a topological ordering of D. Consider the cyclic ordering $O = v_1, \ldots, v_n, v_1$. For every pair of nonadjacent vertices v_i, v_j with $i < j$, add the arc $v_i \to v_j$. Since every arc of D is directed forward in the topological ordering, the resulting oriented graph is a transitive tournament. Moreover, O is a round ordering of this tournament. It follows from Lemma 4.5 that O is an excellent cyclic ordering of D. By Lemma 4.2, the ordering O is also nice. □

We next consider tournaments. Following [3], a cyclic ordering $O = v_1, \ldots, v_n, v_1$ of an oriented graph D is called *round* if, for every i,

$$N_D^+(v_i) = \{v_{i+1}, \ldots, v_{i+d_D^+(v_i)}\} \quad \text{and} \quad N_D^-(v_i) = \{v_{i-d_D^-(v_i)}, \ldots, v_{i-1}\},$$

where indices are read modulo n. A local tournament is called *locally transitive* if the in-neighborhood and the out-neighborhood of every vertex induce transitive tournaments. We shall use the following known characterization.

Theorem 4.7 ([3, Theorem 3])**.** *A connected oriented graph D has a round ordering if and only if D is a locally transitive local tournament. Furthermore, there is a polynomial time algorithm which decides whether a given oriented graph is round and, if so, finds a round ordering.*

The fact that every round ordering is nice follows directly from the definitions. We prove that the converse holds for tournaments.

Proposition 4.8. *Let T be a tournament and let O be a cyclic ordering of $V(T)$. Then O is nice if and only if O is round.*

Proof. Suppose first that O is round. Let $x \to y \to z$ be a directed path in T. Since $x \to y$, the vertex y lies in the consecutive out-neighbor interval immediately after x. Suppose that $\mathrm{Cyc}_O(x, y, z)$ does not hold. Then, starting from x, one meets z before y. By roundness at x, this implies $x \to z$.

Since $y \to z$, the vertex z lies in the consecutive out-neighbor interval immediately after y. Moreover, starting from y, one meets x before z. Roundness at y therefore implies $y \to x$, contradicting $x \to y$. Thus $\mathrm{Cyc}_O(x, y, z)$ holds for every directed path $x \to y \to z$. By Lemma 4.1, O is nice.

Conversely, suppose that O is nice. Fix a vertex $r \in V(T)$, and write the remaining vertices in clockwise order after r as $w_1, \ldots, w_{n-1}$. We claim that no in-neighbor of r occurs before an out-neighbor of r. Suppose, to the contrary, that there are indices $a < b$ such that

$$w_a \to r \qquad \text{and} \qquad r \to w_b.$$

Since T is a tournament, exactly one of $w_a \to w_b$ and $w_b \to w_a$ holds.

If $w_a \to w_b$, then $w_a \to r \to w_b$ is a directed path. By Lemma 4.1, niceness requires $\mathrm{Cyc}_O(w_a, r, w_b)$. However, starting from w_a, one meets w_b before r, a contradiction.

If $w_b \to w_a$, then $r \to w_b \to w_a$ is a directed path. By Lemma 4.1, niceness requires $\mathrm{Cyc}_O(r, w_b, w_a)$. However, starting from r, one meets w_a before w_b, again a contradiction.

Therefore all out-neighbors of r occur before all in-neighbors of r in the clockwise order after r. Hence the out-neighbors of r occur consecutively immediately after r, while its in-neighbors occur consecutively immediately before r. Since r was arbitrary, O is round. □

Corollary 4.9. *Nice cyclic ordering recognition is polynomial time solvable on tournaments.*

Proof. A tournament with at least two vertices is connected. By Proposition 4.8, a tournament has a nice cyclic ordering if and only if it has a round ordering. The result therefore follows from Theorem 4.7. □

Proof of Theorem 1.4. The equivalence between niceness and roundness is exactly Proposition 4.8. The polynomial time recognition statement is Corollary 4.9. □

4.3 A nice ordering need not imply an excellent ordering

We now prove Theorem 1.5.

Let D be the oriented graph with vertex set $\{0, 1, 2, 3, 4, 5, 6, 7\}$ and arc set

$$A(D) = \{0 \to 1,\ 0 \to 5,\ 7 \to 0,\ 4 \to 1,\ 6 \to 1,\ 7 \to 1,\ 2 \to 4,\ 6 \to 2,$$
$$3 \to 4,\ 3 \to 6,\ 4 \to 5,\ 4 \to 6,\ 5 \to 6,\ 6 \to 7\}.$$

The graph D is shown in figure 1, with the vertices placed according to O_0.

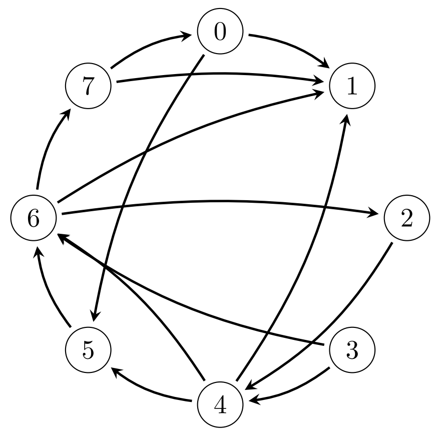

Figure 1: The oriented graph D. The vertices are placed clockwise according to the cyclic ordering $O_0 = 0, 1, 2, 3, 4, 5, 6, 7, 0$.

We shall show that D has a nice cyclic ordering but no excellent cyclic ordering.

We first verify that the displayed ordering is nice.

Lemma 4.10. *The cyclic ordering $O_0 = 0, 1, 2, 3, 4, 5, 6, 7, 0$ is nice.*

Proof. By Lemma 4.1, it is enough to check every directed path of length two. For a path $x \to v \to z$, the condition $\mathrm{Cyc}_{O_0}(x, v, z)$ is equivalent to saying that, when moving clockwise from v, the out-neighbor z is encountered before the in-neighbor x. This is immediate from the following table:

Thus every directed path of length two has the required cyclic order, and O_0 is nice. □

We then show that it is the only nice cyclic ordering up to rotation.

Lemma 4.11. *Every nice cyclic ordering of D is O_0, up to cyclic rotation.*

Proof. Let O be a nice cyclic ordering of D. By Lemma 4.1, every directed path of length two in D must occur in the corresponding cyclic order. We prove the uniqueness of O in three steps.

Claim 1. $\mathrm{Cyc}_O(6, 7, 0, 1)$.

Proof of Claim 1. The directed paths $6 \to 7 \to 1$ and $7 \to 0 \to 1$ give $\mathrm{Cyc}_O(6, 7, 1)$ and $\mathrm{Cyc}_O(7, 0, 1)$.

The first relation says that, starting from 6, one meets 7 before 1. The second says that, starting from 7, one meets 0 before 1. Hence the vertices $6, 7, 0, 1$ occur in this cyclic order. This proves Claim 1. ■

Table 1: Verification that the cyclic ordering $O_0 = 0, 1, 2, 3, 4, 5, 6, 7, 0$ is nice.

v	$N_D^+(v)$	$N_D^-(v)$	Clockwise check from v
0	$\{1,5\}$	$\{7\}$	$1,5$ occur before 7
1	$\varnothing$	$\{0,4,6,7\}$	Vacuous
2	$\{4\}$	$\{6\}$	4 occurs before 6
3	$\{4,6\}$	$\varnothing$	Vacuous
4	$\{1,5,6\}$	$\{2,3\}$	$5,6,1$ occur before $2,3$
5	$\{6\}$	$\{0,4\}$	6 occurs before $0,4$
6	$\{1,2,7\}$	$\{3,4,5\}$	$7,1,2$ occur before $3,4,5$
7	$\{0,1\}$	$\{6\}$	$0,1$ occur before 6

Claim 2. $\mathrm{Cyc}_O(4, 5, 6, 7, 0, 1, 2)$.

Proof of Claim 2. The directed path $4 \to 6 \to 1$ gives $\mathrm{Cyc}_O(4, 6, 1)$. By Claim 1, the interval $(6, 1)_O$ contains 7 and 0 in this order. If $4 \in (6, 1)_O$, then starting from 4 one would meet 1 before 6, contradicting $\mathrm{Cyc}_O(4, 6, 1)$. Thus $4 \notin (6, 1)_O$, and therefore $\mathrm{Cyc}_O(4, 6, 7, 0, 1)$. Next, the directed path $4 \to 5 \to 6$ gives $\mathrm{Cyc}_O(4, 5, 6)$. Hence $5 \in (4, 6)_O$, and so $\mathrm{Cyc}_O(4, 5, 6, 7, 0, 1)$.

Finally, the directed path $2 \to 4 \to 1$ gives $\mathrm{Cyc}_O(2, 4, 1)$. Since $\mathrm{Cyc}_O(4, 5, 6, 7, 0, 1)$, the interval $(4, 1)_O$ contains $5, 6, 7, 0$. If $2 \in (4, 1)_O$, then starting from 2 one would meet 1 before 4, contradicting $\mathrm{Cyc}_O(2, 4, 1)$. Hence $2 \notin (4, 1)_O$, so $2 \in (1, 4)_O$. Consequently, $\mathrm{Cyc}_O(4, 5, 6, 7, 0, 1, 2)$. This proves Claim 2. ■

Claim 3. The vertex 3 lies in the interval $(2, 4)_O$.

Proof of Claim 3. The directed paths $3 \to 4 \to 6$ and $3 \to 6 \to 2$ give $\mathrm{Cyc}_O(3, 4, 6)$ and $\mathrm{Cyc}_O(3, 6, 2)$.

The first relation implies that $3 \notin (4, 6)_O$, and hence $3 \in (6, 4)_O$. The second relation implies that $3 \notin (6, 2)_O$, and hence $3 \in (2, 6)_O$. By Claim 2, the intersection $(6, 4)_O \cap (2, 6)_O$ is exactly the interval $(2, 4)_O$. Therefore $3 \in (2, 4)_O$. This proves Claim 3. ■

By Claim 2 and Claim 3, the vertices occur in the cyclic order $\mathrm{Cyc}_O(4, 5, 6, 7, 0, 1, 2, 3)$, equivalently $\mathrm{Cyc}_O(0, 1, 2, 3, 4, 5, 6, 7)$. Since these are all vertices of D, the cyclic ordering O is exactly $O_0 = 0, 1, 2, 3, 4, 5, 6, 7, 0$, up to cyclic rotation. □

Proof of Theorem 1.5. By Lemma 4.10, the graph D has a nice cyclic ordering. Suppose that D has an excellent cyclic ordering. By Lemma 4.2, this ordering is nice, and hence, by Lemma 4.11, it is O_0 up to cyclic rotation.

However, O_0 is not excellent. Indeed, the arcs $0 \to 5$ and $4 \to 1$ have their endpoints in the cyclic order $0, 1, 4, 5$, which is the forbidden pattern v_i, v_t, v_s, v_j, with $v_i = 0, v_t = 1, v_s = 4, v_j = 5$. Thus no excellent cyclic ordering of D exists. □

5 Concluding remarks

We close with two questions suggested by the results above.

First, Theorem 1.2 shows that acyclic π-linkage completion is NP-complete even for one terminal pair. The reduction does not preserve chordality of the underlying graph, whereas the polynomial-time algorithm for acyclic in-tournament completion is based on a perfect-elimination-ordering description. It is therefore natural to ask whether chordal structure restores tractability.

Problem 5.1. *Determine the complexity of acyclic π-linkage completion for partially oriented graphs whose underlying graph is chordal. Is the problem polynomial-time solvable for fixed k on chordal graphs, interval graphs, or proper interval graphs?*

Second, the example in Section 4 shows that nice cyclic orderings and excellent cyclic orderings are distinct notions. A structural description of this gap remains open.

Problem 5.2. *Characterize the oriented graphs which admit a nice cyclic ordering but no excellent cyclic ordering. Equivalently, determine the minimal obstructions to the implication from niceness to excellence.*

Declaration on the use of generative AI.

The authors used ChatGPT 5.5 Pro for language editing, improving the exposition, and developing proof ideas and preliminary arguments for several key lemmas. All AI-assisted content was carefully checked and revised by the authors, who take full responsibility for the final manuscript.

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